

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

data = pd.read_csv('twitter_training.csv')
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 74681 entries, 0 to 74680
Data columns (total 4 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   2401                  74681 non-null  int64
 1   Borderlands           74681 non-null  object
 2   Positive              74681 non-null  object
 3   im getting on borderlands and i will murder you all , 73995 non-null  object
dtypes: int64(1), object(3)
memory usage: 2.3+ MB
```

```
data.columns = ['Id', 'Entity', 'Sentiment', 'Tweet']
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 74681 entries, 0 to 74680
Data columns (total 4 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   Id                    74681 non-null  int64
 1   Entity                74681 non-null  object
 2   Sentiment             74681 non-null  object
 3   Tweet                73995 non-null  object
dtypes: int64(1), object(3)
memory usage: 2.3+ MB
```

```
data.isnull().sum()
```

```

      0
Id      0
Entity  0
Sentiment  0
Tweet  686
dtype: int64
```

```
data.dropna(inplace=True)
```

```
data.isnull().sum()
```

```

      0
Id      0
Entity  0
Sentiment  0
Tweet  0
dtype: int64
```

```
data.duplicated().sum()
```

```
2340
```

```
data.drop_duplicates(inplace=True)
```

```
data.duplicated().sum()
```

```
0
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 71655 entries, 0 to 74680
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0    Id          71655 non-null  int64
1    Entity      71655 non-null  object
2    Sentiment   71655 non-null  object
3    Tweet       71655 non-null  object
dtypes: int64(1), object(3)
memory usage: 2.7+ MB
```

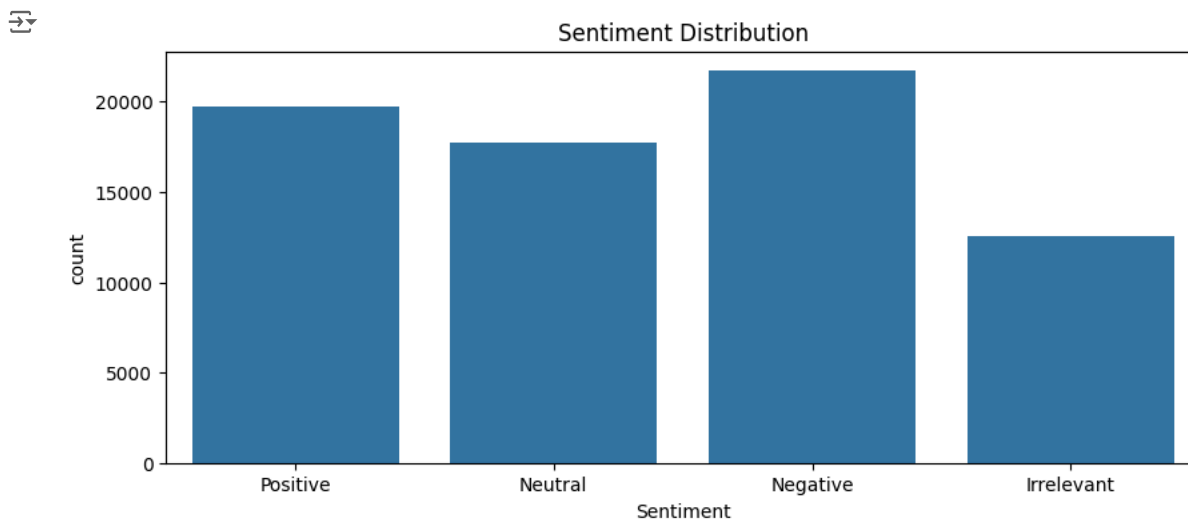
```
data.head()
```

	Id	Entity	Sentiment	Tweet
0	2401	Borderlands	Positive	I am coming to the borders and I will kill you...
1	2401	Borderlands	Positive	im getting on borderlands and i will kill you ...
2	2401	Borderlands	Positive	im coming on borderlands and i will murder you...
3	2401	Borderlands	Positive	im getting on borderlands 2 and i will murder ...
4	2401	Borderlands	Positive	im getting into borderlands and i can murder v...

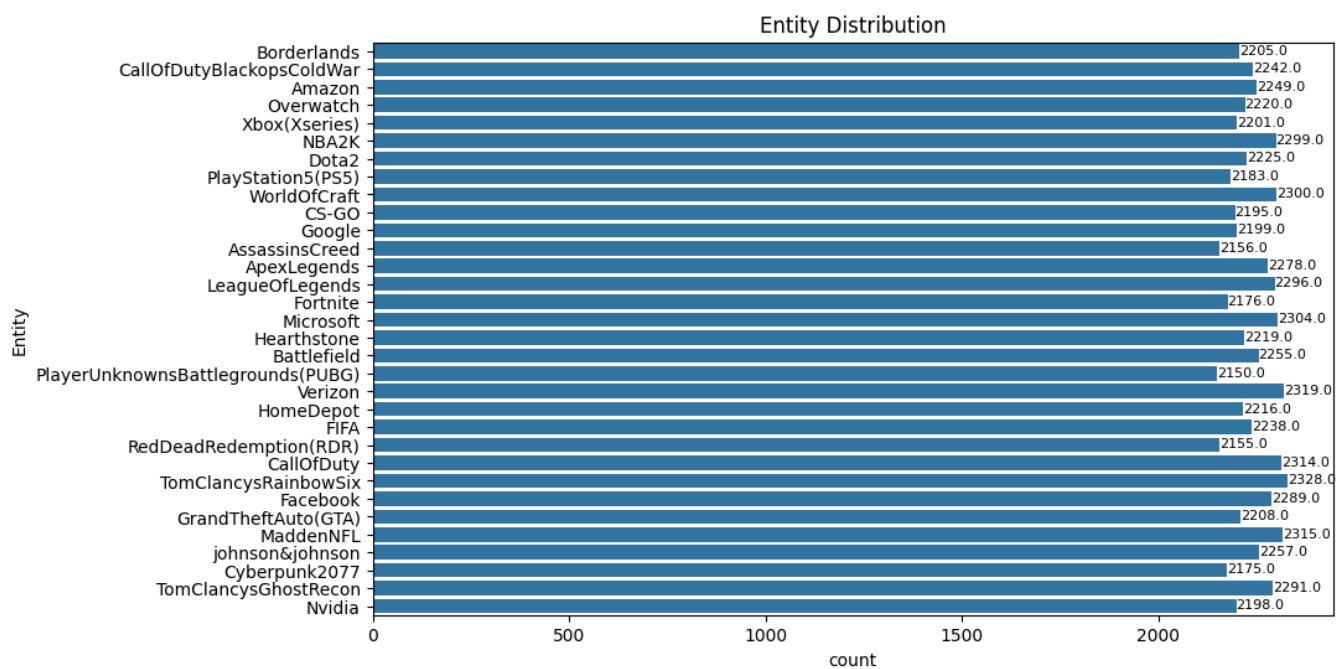
Next steps:

[Generate code with data](#)[View recommended plots](#)[New interactive sheet](#)

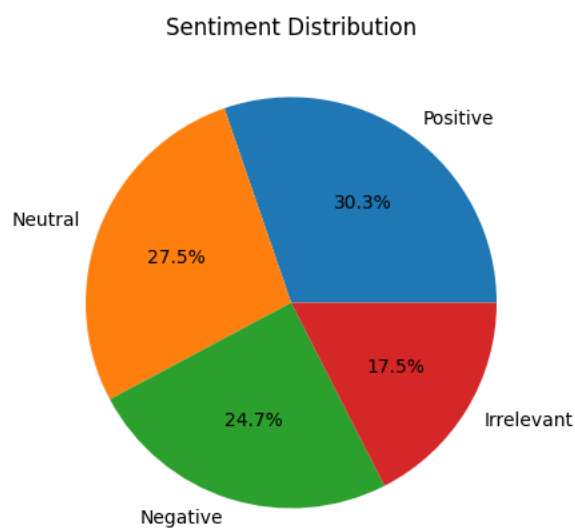
```
plt.figure(figsize=(10, 4))
sns.countplot(x='Sentiment', data=data)
plt.title('Sentiment Distribution')
plt.show()
```



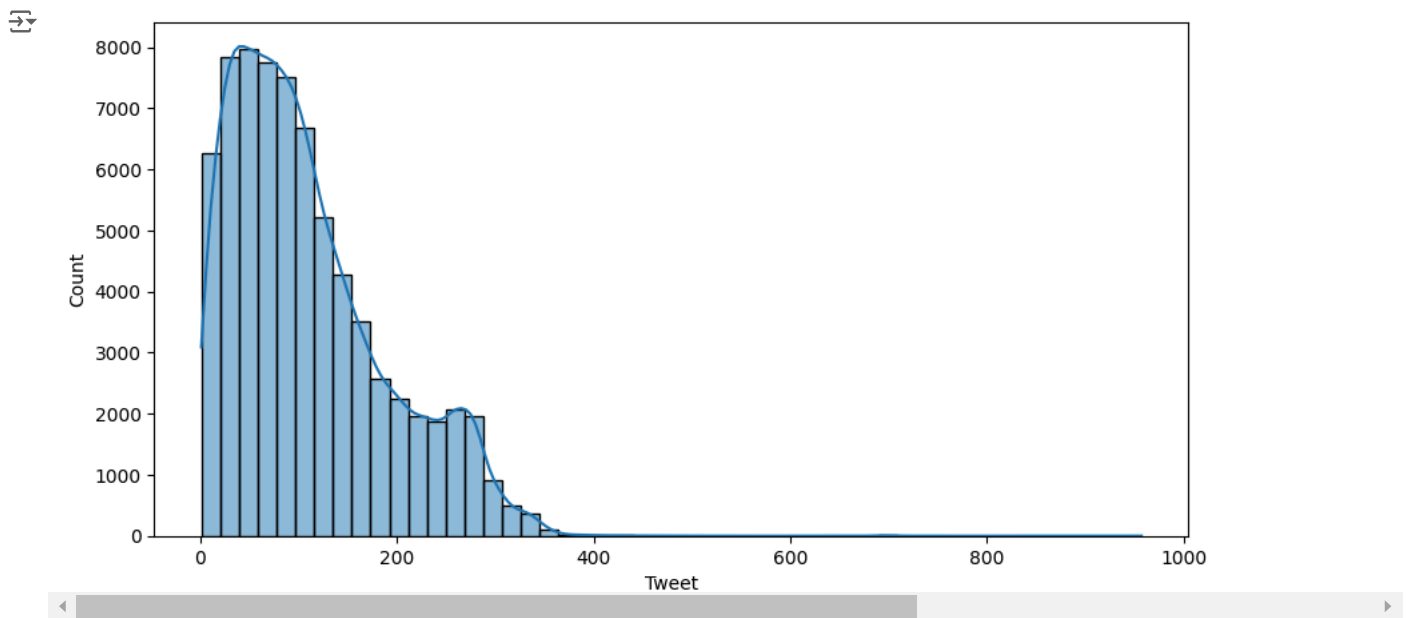
```
plt.figure(figsize=(10, 6))
sns.countplot(y='Entity', data=data)
for bar in plt.gca().patches:
    plt.gca().annotate(f'{bar.get_width()}', (bar.get_x() + bar.get_width(), bar.get_y()), size=8, ha='left', va='top')
plt.title('Entity Distribution')
plt.show()
```



```
plt.figure(figsize=(10, 5))
plt.pie(data['Sentiment'].value_counts(), labels=data['Sentiment'].unique(), autopct='%1.1f%%')
plt.title('Sentiment Distribution')
plt.show()
```



```
plt.figure(figsize=(10, 5))
sns.histplot(data['Tweet'].str.len(), bins=50, kde=True)
plt.show()
```



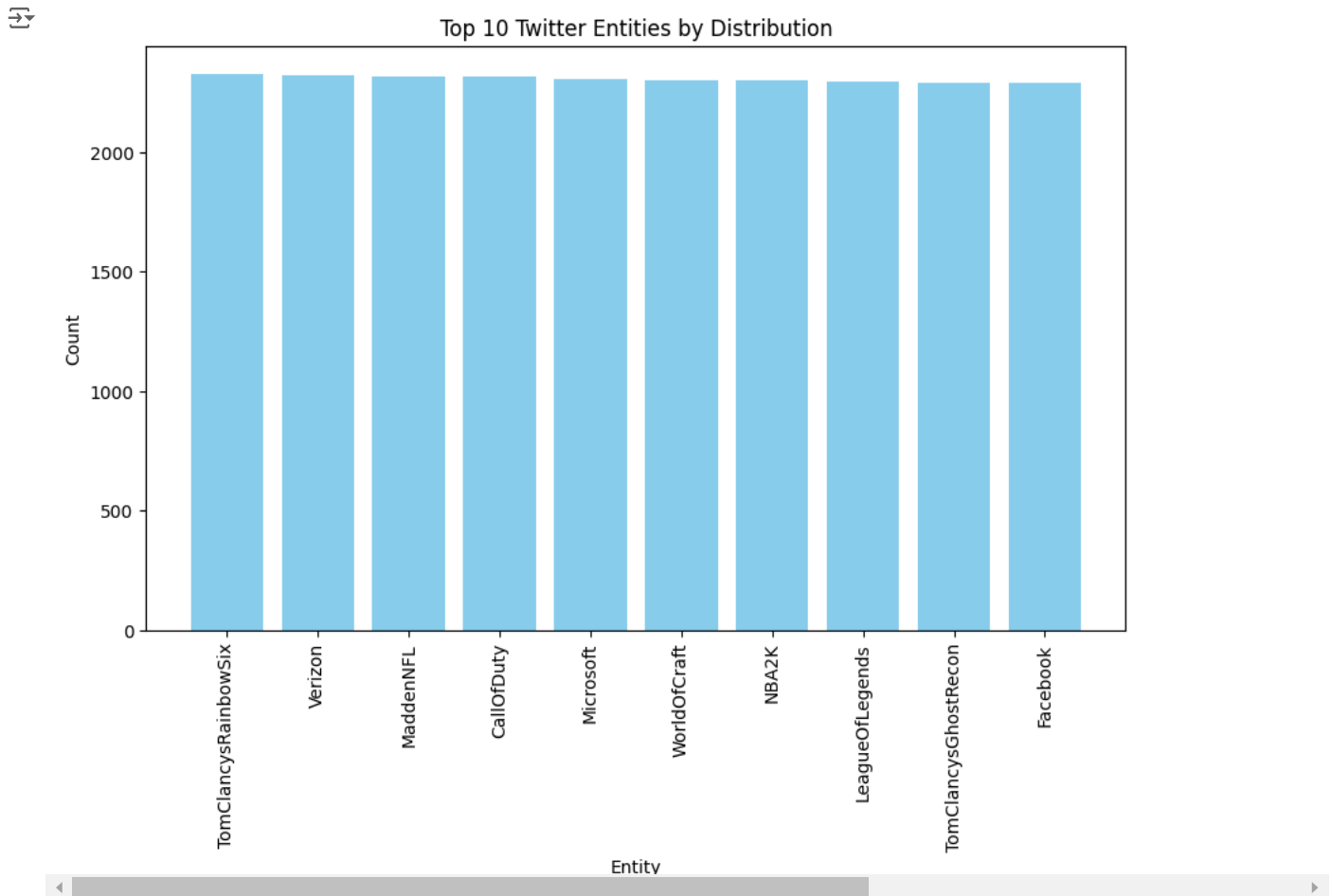
```
entity_distribution = data['Entity'].value_counts()

top_10_entities = entity_distribution.head(10)

plt.figure(figsize=(10,6))
plt.bar(top_10_entities.index, top_10_entities.values, color='skyblue')
plt.title('Top 10 Twitter Entities by Distribution')
plt.xlabel('Entity')
plt.ylabel('Count')

plt.xticks(rotation=90)

plt.show()
```



```
plt.figure(figsize=(10,5))
sns.countplot(data=data, x='Sentiment', hue='Entity')
```



```
wordcloud = WordCloud(width=800, height=400, background color='white').generate(' '.join(data['Tweet']))
```

